Aashish Gupta

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PROFESSIONAL APPOINTMENTS

- **VICO postdoctoral fellow at the University of Virginia (UVA), USA**
  2025-present; *Advisor: Dr. Ilse Cleeves*
  I am leading multiple research projects to characterize the impact of star-forming environments on planet-forming disks.

- **Doctoral researcher at European Southern Observatory (ESO), Germany**
  2021-2024; *Advisor: Dr. Anna Miotello*
  I demonstrated that planet-forming disks often interact with surrounding molecular clouds, and such clouds can drastically enhance their mass-budget to form planets.

- **Graduate researcher at National Central University (NCU), Taiwan**
  2019-2021; *Advisor: Dr. Chen, Wen-Ping*
  I diagnosed the spatial and kinematic distribution of the young stars with respect to molecular clouds in the Rho Ophiuchus star-forming region, providing new insights into star cluster formation.

- **Summer intern at Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taiwan**
  2020; *Advisor: Dr. Yen, Hsi-Wei*
  Using data from JCMT and SMA, I investigated correlations between magnetic fields and gas kinematics for protostars in Perseus molecular clouds.

- **Project-based internship at Aryabhatta Research Institute of Observational Sciences (ARIES), India**
  2018; *Advisor: Dr. Jeewan Pandey*
  I developed a pipeline to detect and analyze stellar flares observed in *Kepler*'s lightcurves.

- **Summer intern at Aryabhatta Research Institute of Observational Sciences (ARIES), India**
  2017; *Advisor: Dr. Sneh Lata*
  I developed a pipeline to analyse time series data of RR Lyrae Variables and estimate their physical parameters.

- **Undergraduate researcher at Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, India**
  2016; *Advisor: Prof. Puneet Tandon*
  I assisted in the design and fabrication of a machine for 3D printing objects out of sugar.

EDUCATION

Year	Qualification	Institute	Grade
2021-present	Doctor of Philosophy (Astronomy)	European Southern Observatory and Ludwig Maximilian University, Germany	1.2; Magna cum laude
2019-2021	Master of Science (Astronomy)	National Central University, Taiwan	89.6/100
2015-2019	Bachelor of Technology (Mechanical Engineering)	Indian Institute of Information Technology, Design and Manufacturing, Jabalpur, India	8.1/10

Besides these, I have done 15 online courses related to astrophysics, astronomy, data science, and machine learning on online platforms coursera.org, edx.org, and kaggle.com.

SKILLS

- *Programming Languages:* Python, C/C++, Octave, HTML, Shell scripting
- *Softwares:* LaTeX (Typesetting), CASA (Radio data processing), IRAF (Optical data reduction), Git (Software management), DS9 (Imaging), CARTA (Radio imaging)
- *Languages:* Hindi (Native), English (Proficient, IELTS score 8), German (Basic, A1), Spanish (Basic, A1), Mandarin Chinese (Basic speaking and listening)

PUBLICATIONS

Under review

- **A tale of three tails: A misaligned streamer and mysterious structures around [BHB2007]1**
  Gupta, A., Hales, A. S., Cleeves, L. I., Alves, F., Bhowmik, T., Cuello, N., Girart, J. M., Li, Z.-Y., Miotello, A., Zhu, Z., and Zurlo, A.

First-author publications:

- **TIPSY: Trajectory of Infalling Particles in Streamers around Young Stars. Dynamical analysis of streamers around S CrA and HL Tau**
  Gupta, A., Miotello, A., Williams, J. P., Birnstiel, T., Küffmeier, M., Yen, H.-W. (2024), A&A, 683, A133. doi.org/10.1051/0004-6361/202348007
- **Reflections on nebulae around young stars: A systematic search for late-stage infall of material onto Class II disks**
  Gupta, A., Miotello, A., Manara, C. F., Williams, J. P., Facchini, S., Beccari, G., Birnstiel, T., Ginski, C., Hacar, A., Küffmeier, M., Testi, L., Tychoniec, L., Yen, H.-W. (2023), A&A, 670, L8. doi.org/10.1051/0004-6361/202245254
- **Effects of Magnetic Field Orientations in Dense Cores on Gas Kinematics in Protostellar Envelopes**
  Gupta, A., Yen, H.-W., Koch, P., Bastien, P., Bourke, T. L., Chung, E. J., Hasegawa, T., Hull, C. L. H., Inutsuka, S., Kwon, J., Kwon, W., Lai, S.-P., Lee, C. W., Lee, C.-F., Pattle, K., Qiu, K., Tahani, M., Tamura, M., Ward-Thompson, D. (2022), ApJ, 930, 67. doi.org/10.3847/1538-4357/ac63bc
- **Interplay between Young Stars and Molecular Clouds in the Ophiuchus Star-forming Complex**
  Gupta, A., Chen, W.-P. (2022), AJ, 163, 233. doi.org/10.3847/1538-3881/ac5cc8

Co-author publications:

- **Spatially correlated stellar accretion in the Lupus star-forming region: Evidence for ongoing infall from the interstellar medium**
  Winter, A. J., Benisty, M., … *Gupta, A.*, et al. (2024), A&A, 691, A169. doi.org/10.1051/0004-6361/202452120
- **Discovery of an Accretion Streamer and a Slow Wide-angle Outflow around FU Orionis**
  Hales, A. S., *Gupta, A.*, et al. (2024), ApJ, 966, 96. doi.org/10.3847/1538-4357/ad31a1
- **Anatomy of the Class I protostar L1489 IRS with NOEMA: disk, streamers, outflow(s) and bubbles at 3 mm**
  Tanious, M., Le Gal, R., … *Gupta, A.*, et al. (2024), A&A, 687, A92. doi.org/10.1051/0004-6361/202348785
- **A dusty streamer infalling onto the disk of a Class I protostar: ALMA dual-band constraints on grain properties and mass infall rate**
  Cacciapuoti, L., Macías, E., *Gupta, A.*, et al. (2023), A&A, 682, A61. doi.org/10.1051/0004-6361/202347486
- **The VLT MUSE NFM view of outflows and externally photoevaporating discs near the Orion Bar**
  Haworth, T. J., Reiter, M., … *Gupta, A.*, et al. (2023), A&A, 670, L8. doi.org/10.1093/mnras/stad2581
- **The JCMT Transient Survey: Four-year summary of monitoring the submillimeter variability of protostars**
  Lee, Y.-H., Johnstone, D., … *Gupta, A.*, et al. (2021), ApJ, 920, 119. doi.org/10.3847/1538-4357/ac1679
- **No impact of core-scale magnetic field, turbulence, or velocity gradient on sizes of protostellar disks in Orion A**
  Yen, H.-W., Zhao, B., Koch, P. M., *Gupta, A.* (2021), ApJ, 916, 97. doi.org/10.3847/1538-4357/ac0723
- **VR CCD photometry of variable stars in the globular cluster NGC 4147**
  Lata, S., Pandey, A. K., … *Gupta, A.*, et al. (2019), AJ, 158, 51. doi.org/10.3847/1538-3881/ab22a6

PARTICIPATION IN MEETINGS

Invited talks:

- Astronomy Seminar, University of Connecticut, Storrs, USA, October 2025
- Star and Exoplanet Seminar, Yale University, New Haven, USA, October 2025
- Center for Astrophysics | Harvard & Smithsonian, Cambridge, USA, June 2025
- European Astronomical Society Annual Meeting (EAS), Cork, Ireland, June 2025
- StarPlan seminar, University of Copenhagen, Copenhagen, Denmark, December 2024
- Astronomy seminar, University of Exeter, Exeter, UK, February 2024
- Core2disk III workshop, Paris, France, October 2023
- Origins seminar at Chalmers University, Gothenburg, Sweden, August 2023
- ESO star and planet formation seminar, Garching, Germany, February 2023
- ECOGAL seminar, Online, February 2023
- MPE-CAS journal club on star and planet formation, Garching, Germany, July 2022

Contributed talks:

- Born in Fire: Eruptive Stars and Planet Formation, Santiago, Chile, September 2024
- New Heights in Planet Formation, Garching, Germany, July, 2024
- European Astronomical Society Annual Meeting (EAS), Krakow, Poland, July 2023
- Meeting of ALMA Young Astronomers (MAYA), Online, March 2023
- Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan, September 2020

Posters:

- Origins of Solar Systems (Gordon Research Conference), South Hadley, Massachusetts, USA, June 2025
- Inter+Stellar: Harnessing the Intersection Between Stars and the Interstellar Medium, Baltimore, USA, May 2025
- Star@Lyon conference, Lyon, France, June 2023
- Protostars and Planets VII, Kyoto, Japan, April 2023
- Annual Meeting of Astronomy Society of the Republic of China (ASROC), Taipei, Taiwan, September 2020
- Annual Meeting of the Physical Society of Taiwan (TPS), Pingtung, Taiwan, February 2020
- Exploring the Universe: Near Earth space science to extragalactic astronomy (EXPUNIV2018), Kolkata, India, Nov 2018

AWARDS AND ACHIEVEMENTS

- VICO postdoctoral fellowship at UVA, USA
- DFG funding for Ph.D. at ESO, Germany
- MOST scholarship for master's program at NCU, Taiwan
- Award for academic performance at NCU, Taiwan
- Certificate of Merit for academic excellence at IIITDM Jabalpur, India
- Invitation to write an SMA Newsletter article
- Poster presentation award in TPS 2020
- Won several Astronomy competitions like quizzes and hackathons

OBSERVING EXPERIENCE

- PI of ALMA projects:
    - "Zooming out: Disentangling the origins of large-scale structures around planet-forming disks" (project code: 2024.1.00524.S)
    - "Paradigm Shift or Rare Events: Testing the Frequency of Late Infall Using Herbig Disks" (project code: 2025.1.00197.S)
    - "Zooming Out and Peering In: The First Statistical Study of Infall onto Class II Disks " (project code: 2025.1.00328.S)
    - "Is infall of material misaligning planet-forming disks?" (project code: 2025.1.00803.S)
- PI of VLT project "Exploring effects of infall of material onto Class II systems" (project code: 113.26GY)
- PI of SMA project "In or Out: Spectral-Line Survey of a Streamer" (project code: 2025A-S031)
- Member of ALMA large programs DECO (project code: 2022.1.00875.L) and CHEER (project code: 2024.1.01001.L)
- CoI on accepted ALMA, JWST, VLT, SMA, and APEX programs.
- Assisted in optical observations at ARIES, India and Lulin Observatory, Taiwan

MENTORING EXPERIENCE

- Co-supervised Andres Zuleta for "Cool Frontiers: Exploring Dust and Ice in the Cosmos" at UVA (2025)
- Co-supervised Lauren Mason for the ESO Summer Research Programme (2024)
- Supervised Jayden Burg, a high school student, for one week internship (Schülerpraktikum) (2024)
- Co-supervised Alessandro Ruzza for the ESO Summer Research Programme (2023)
- Academic Helper for Physics and Mathematics at IIITDMJ Counseling Services (2016-2018)
- Student guide for the first year students at IIITDMJ (2016)

ACADEMIC SERVICE

- Chair of Community Building Committee of the UVA Postdoc Association (2025-present)
- Local Organizing Committee member for "Cool Frontiers: Exploring Dust and Ice in the Cosmos" at UVA (2025)
- Local Organizing Committee member for DECO All Team meeting at ESO, Germany (2024)
- Local Organizing Committee member for "New Heights in Planet Formation" at ESO, Germany (2024)
- Scientific Assistant for ESO Observing Programmes Committee (2024)
- Member of selection committee for ESO Summer Research Programme (2024)
- Co-organizer of Wine & Cheese seminars at ESO, Germany (2023-2024)
- Chaired for Hypatia Colloquium at ESO, Germany (2023)
- Refereed for Science Advances and The Astrophysical Journal

OUTREACH

- Organizing Committee member for "Girls Exploring the Universe" at Charlottesville, USA (2025)
- Co-organizer of Astronomy for Non-astronomers talks at ESO, Germany (2024)
- Coordinator of The Astronomy and Physics Society of IIITDM Jabalpur, India. We organized events like telescope nights, exhibitions, quizzes, etc. to promote astronomy among a variety of audiences. (2016-2019)
- E-Astronomer at RAD@home Astronomy Collaboratory, an Indian citizen-science research programme (2018-2019)